013

LCD FOR Z80-DRIVEN COMPUTERS

There is a growing tendency to use liquid-crystal displays (LCD) as the screen of computer monitors. Such displays may also be used where the normal monitor is too large or draws too much current; they are readily available. An LCD is normally driven by a microprocessor: in the proposed circuit by a Z80.

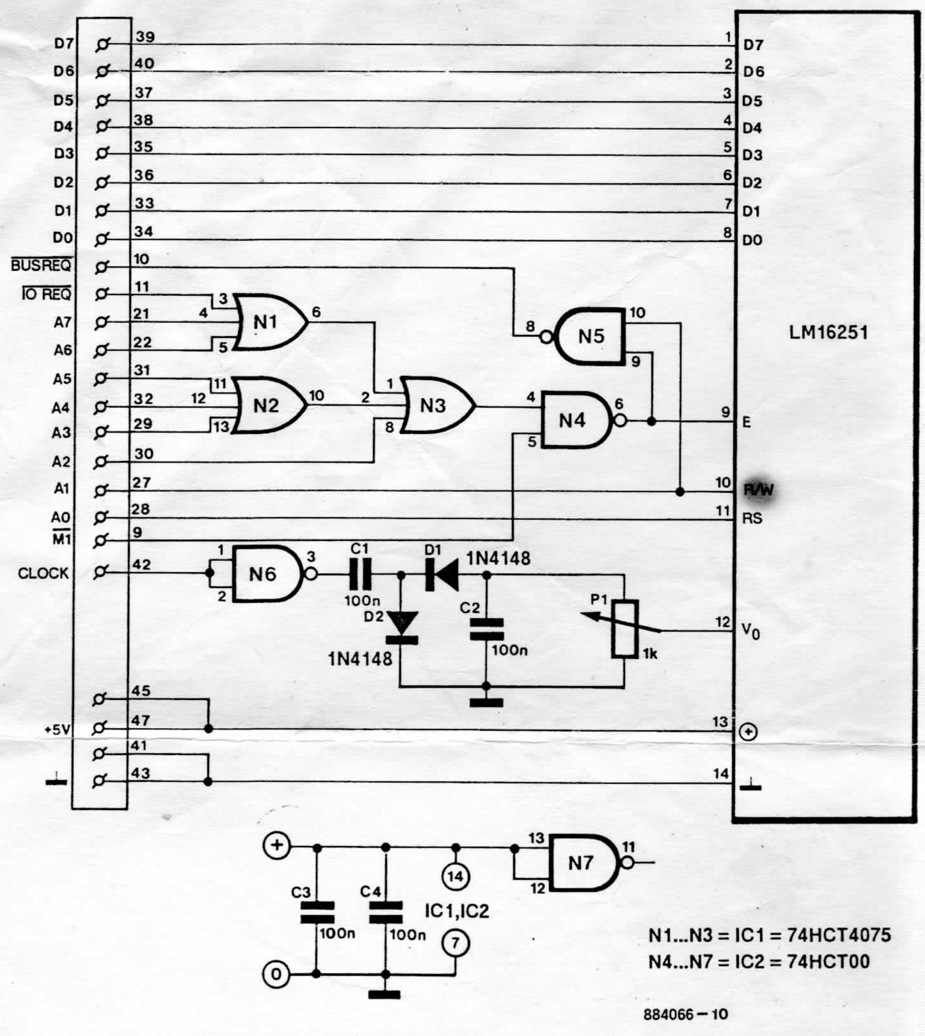
The display in the proposed circuit is a Sharp Type LM16251: a full description of this appeared in the May 1986 issue of this magazine. It is located in the I/O region, addresses 0 to 3, of the processor. This arrangement enables the circuit also to be used in combination with the 32-bit I/O and timer cartridge described in the January 1987 issue of this magazine. This cartridge does not use the lowest four addresses (choose address 0 for the cartridge so that an additional I/O region of 0 to 15 is obtained).

The address coding is effected by gates N_1 to N_4. When A_2 to A_7 are, and IO REQ becomes low, the output of N_3 goes low. If $\overline{M1}$ is high (no interrupt demanded), N_4 outputs a 1 and an enable signal is given to the display.

Depending on the logic levels at inputs R/W and RS, data is transmitted or received. The RD and WR outputs of the Z80 are not used, because the R/W and RS signals of the LM16251 must be stable not later than 140 ns before the E input goes high. If the RD or WR signals of the processor were used, the E input of the display would be accessed together with the other signals and that is not permitted.

By using an address line, the timing is arranged by that of the Z80, because the address bus must be stable not later than 320 ns (180 ns for Z80A) before an IO REQ signal is generated. Owners of a Z80B-driven computer might have some problems here because the time delay is then only 110 ns. Note that MSX computers invariably use a Z80A.

The negative voltage for the contrast setting (P_1) of the display is provided by N_6. Note that some types of display need a positive voltage for the contrast setting. Wire link 'a' provides a negative supply, and 'b' a positive one. Link 'a' is required for the LM16251. If another type of display is used, make sure that the pin numbering is the same as shown in the diagram.



Gate N_5 serves to render the $\overline{BUSDIR}$ line low at an I/O read command in MSX systems. In other systems, this gate is not required.